

CPS I — Exercise Sheet 2

1	2	3	Σ
3/3	4/4	8/9	15/16

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1. Implications

3/3

Group 1:

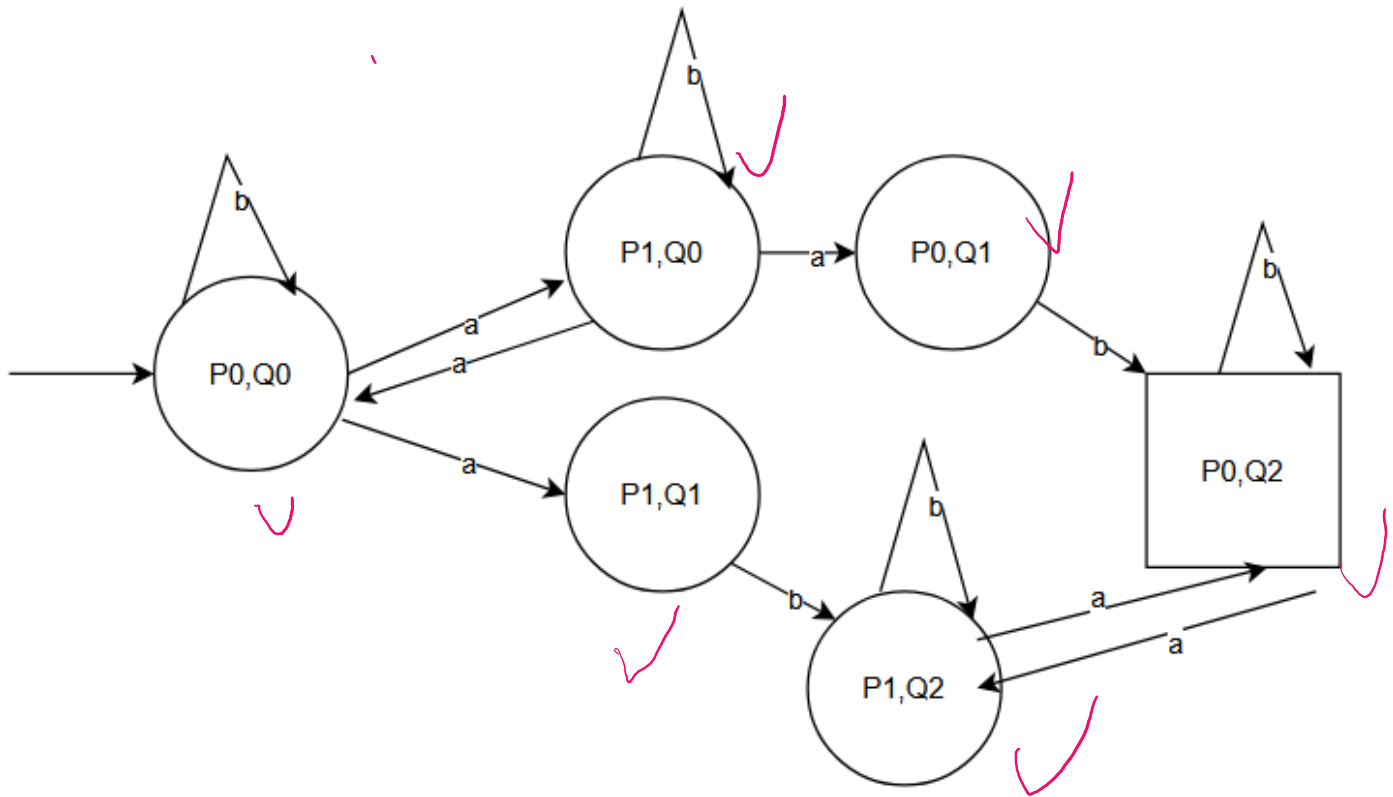
- A implies B ✓
- A is a sufficient condition for B ✓
- A only if B ✓
- A is stronger than B ✓
- $A \wedge \neg B$ does not hold ✓
- $\neg A \vee B$ holds ✓
- If not B, then not A ✓

Group 2:

- B implies A ✓
- A is a necessary condition for B ✓
- A if B ✓
- A is weaker than B ✓

2. Intersection of Finite Automata

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3. Buchi Automata

8/9

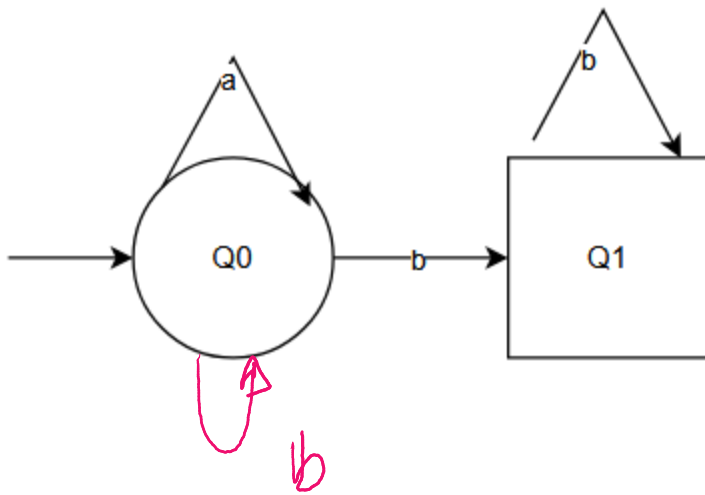
1. Define

$$L_1 = \{x_0x_1 \dots \mid (\forall i \in \mathbb{N}_0. x_i \in \Sigma) \text{ and } \exists j \in \mathbb{N}_0. \forall i \in \mathbb{N}_0. i > j \text{ and } x_i \neq a\}$$

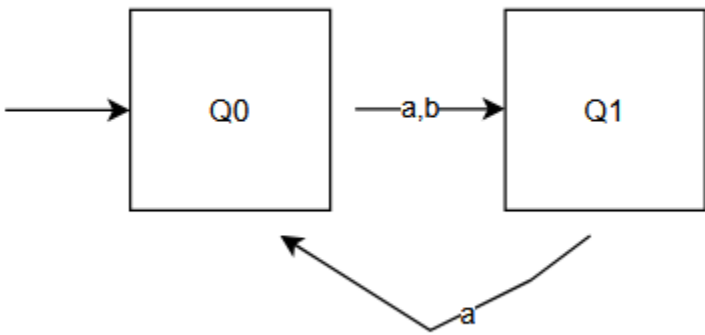
$$L_2 = \{x_0x_1 \dots \mid (\forall i \in \mathbb{N}_0. x_i \in \Sigma) \text{ and } \forall i \in \mathbb{N}_0. x_{2i+1} = a\}$$

2. Buchi Automata

⇒ - 0.5



does not accept, for
example $(ab)^5 b^w$
which is in L_1 .
— 0.5



• L1

- $Q: \{Q_0, Q_1\}$
- $\Sigma: \{a, b\}$
- $\delta: \begin{cases} (Q_0, a, Q_0), (Q_0, b, Q_1) \\ (Q_1, b, Q_1) \end{cases}$
- $Q_{\text{init}}: \{Q_0\}$
- $F: \{Q_1\}$

• L2

- $Q: \{Q_0, Q_1\}$
- $\Sigma: \{a, b\}$
- $\delta: \begin{cases} (Q_0, a, Q_1), (Q_0, b, Q_1) \\ (Q_1, a, Q_0) \end{cases}$
- $Q_{\text{init}}: \{Q_0\}$
- $F: \{Q_0, Q_1\}$